Software Requirements Specification

for

Dynamic Expiry-Based Grocery & Zero-Waste Marketplace

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Rayyan	07/30/2026	Initial draft based on Vision & Scope discussion	0.1 draft
Darsh	08/01/2026	Baseline approved after internal review	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for Release 1.0 of the Dynamic Expiry-Based Grocery & Zero-Waste Marketplace (referred to throughout this document as "the system" or "the COS-style platform"). It is intended for use by the developers, testers, project managers, and academic evaluators who design, implement, verify, and assess the system. Unless otherwise indicated, every requirement stated in this document is considered committed for Release 1.0 and is written to be unambiguous, verifiable, and traceable to a unique identifier.

1.2 Scope

The Dynamic Expiry-Based Grocery & Zero-Waste Marketplace is a multi-vendor e-commerce web and mobile platform that allows partner grocery stores to list surplus or near-expiry inventory for sale at automatically reduced prices. An hourly decay-based pricing engine progressively lowers the price of each listed item as its expiry date and time approach, incentivizing customers to purchase food that would otherwise be discarded. The platform connects four classes of user — Customer, Grocery Store Owner, Delivery Partner, and Administrator — and provides registration, product listing, inventory and expiry tracking, dynamic pricing, search, cart, secure payment, order management, notifications, food-waste analytics, and platform administration.

Release 1.0 covers the core marketplace transaction loop: account registration and login; store onboarding and verification; product listing, inventory, and expiry tracking; the hourly dynamic pricing engine; search and filtering; cart and checkout; payment processing; order and delivery management; notifications; a food-waste analytics dashboard; and the administrative back office. Release 1.0 explicitly excludes: direct point-of-sale (POS) system integration for partner stores; a native loyalty/rewards program; multi-country/multi-currency operation; and subscription-based recurring grocery orders. These items are candidates for later releases and are recorded in Appendix C-equivalent tracking as future scope, not as requirements of this document.

1.3 Intended Audience

This document is intended for the following readers:

- Software developers and architects responsible for designing and building the platform.
- Quality assurance engineers responsible for deriving test cases and verifying conformance to the requirements herein.
- Project managers and academic supervisors tracking scope, priority, and delivery progress.
- Grocery store business stakeholders reviewing the functionality being built on their behalf.
- Course evaluators assessing this document as a Software Engineering case study.

Readers unfamiliar with the domain should begin with Section 2 (Overall Description) before proceeding to the detailed feature requirements in Section 4. Readers who only need the itemized "shall" requirements for test-case derivation may proceed directly to Sections 4 through 6.

1.4 References

- 1. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications, Institute of Electrical and Electronics Engineers.
- 2. Dynamic Expiry-Based Grocery & Zero-Waste Marketplace — Vision and Scope Document, Version 1.0.

- 3. PCI Security Standards Council, Payment Card Industry Data Security Standard (PCI-DSS), Version 4.0.
- 4. Government of India, Digital Personal Data Protection Act (DPDP Act), 2023.
- 5. Food Safety and Standards Authority of India (FSSAI), Packaging and Labelling Regulations, 2020.
- 6. Platform UI/UX Design and Accessibility Guidelines, Version 1.0 (internal).

2. Overall Description

2.1 Product Perspective

The Dynamic Expiry-Based Grocery & Zero-Waste Marketplace is a new, self-contained, multi-vendor system; it is not a follow-on to an existing internal product, though it competes conceptually with manual in-store markdown processes and existing surplus-food mobile applications. The system is composed of a customer-facing web and mobile storefront, a store-owner web portal, a delivery-partner mobile application, an administrative back office, and a set of backend services — including the pricing engine, inventory service, order service, notification service, and analytics service — that communicate over REST APIs. The system integrates with external services for payment processing, e-mail/SMS delivery, push notifications, and map-based geolocation. Figure captions and a full context diagram description are provided in Appendix A.

2.2 Product Functions

At a high level, the system shall let users perform the following major functions, each detailed further in Section 4:

- Register and authenticate as a Customer, Grocery Store Owner, Delivery Partner, or Administrator.
- Submit and verify grocery store registrations.
- Create, edit, and manage product listings, including inventory quantities and expiry dates.
- Automatically track product expiry and classify listings by expiry-proximity band.
- Automatically recalculate product prices every hour using a decay-based discount algorithm.
- Search, filter, and sort available discounted products.
- Add products to a cart with a time-limited price hold, and complete secure checkout.
- Process payments and refunds through an integrated payment gateway.
- Manage the full order lifecycle from placement through delivery and rating.
- Send notifications for order updates, price drops, and expiry alerts.
- View food-waste and sales analytics dashboards.
- Administer stores, users, listings, and platform-wide configuration parameters.

2.3 User Classes and Characteristics

• Customer (favored) — price- and sustainability-motivated shoppers who check the marketplace frequently for newly discounted, near-expiry items; ~70% of orders via mobile app.
• Grocery Store Owner — lists surplus stock and fulfills orders, typically at store opening/closing; assumed basic computer literacy, favoring guided workflows and sensible pricing defaults.
• Delivery Partner — store staff or gig couriers who accept and fulfill delivery requests via a lightweight, one-handed-usable mobile app on entry-level Android devices.
• Administrator — a small set (<20) of platform staff verifying stores, configuring rules, and resolving disputes as power users of the admin panel.

2.4 Operating Environment

- OE-1: The system shall operate as a responsive web application supporting the current and prior major versions of Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge.
- OE-2: The system shall provide native mobile applications for Android 10.0+ and iOS 15.0+ for the Customer and Delivery Partner user classes.
- OE-3: The system's backend services shall be deployed on a cloud infrastructure provider supporting containerized, horizontally scalable microservices (e.g., AWS, Azure, or GCP).
- OE-4: The system shall use a relational database management system (e.g., PostgreSQL 14 or later) for transactional data and a caching layer (e.g., Redis) for frequently read data such as current product prices.
- OE-5: The system shall permit access from the public Internet over HTTPS, with no dependency on a private corporate network.

2.5 Design and Implementation Constraints

- CO-1: All payment processing shall comply with PCI-DSS Version 4.0 requirements [3].
- CO-2: The collection, storage, and processing of personal data shall comply with the Digital Personal Data Protection Act, 2023 [4].
- CO-3: Product listings for packaged food items shall display information consistent with FSSAI packaging and labelling regulations [5].
- CO-4: The system architecture shall be organized as independently deployable microservices communicating over versioned REST APIs, to allow the pricing engine to scale independently of other services.
- CO-5: All user interface screens shall conform to the Platform UI/UX Design and Accessibility Guidelines, Version 1.0 [6], including WCAG 2.1 Level AA accessibility targets.
- CO-6: All source code, database schema changes, and infrastructure-as-code shall be maintained under version control with mandatory peer review before merge to the main branch.

2.6 User Documentation

- UD-1: A searchable online help center and an interactive first-time walkthrough (for Store Owners, explaining the decay-pricing engine) shall be provided, with contextual tooltips on pricing configuration.

2.7 Assumptions and Dependencies

- AS-1: It is assumed that partner grocery stores have continuous Internet access and at least one Internet-enabled device available to store staff during operating hours.
- AS-2: It is assumed that Store Owners will enter accurate expiry dates and times for listed products; the system does not independently verify physical product freshness.
- AS-3: It is assumed that Customers have a valid payment method supported by the integrated payment gateway.
- DE-1: The operation of the system depends on the continued availability and uptime of a third-party PCI-DSS compliant payment gateway.
- DE-2: The operation of order notifications depends on a third-party SMS gateway and push-notification service (e.g., Firebase Cloud Messaging / Apple Push Notification service).
- DE-3: The distance-based search filter and delivery-partner routing depend on a third-party mapping and geolocation API.

3. External Interface Requirements

3.1 User Interfaces

- UI-1: The system's screens shall conform to the Platform UI/UX Design and Accessibility Guidelines, Version 1.0 [6].
- UI-2: The system shall provide a context-sensitive help link on every screen.
- UI-3: The system shall display the current expiry-proximity band and dynamically calculated price prominently on every product card and product detail page.
- UI-4: All primary navigation and checkout flows shall be fully operable using keyboard-only input, in addition to touch and mouse input.
- UI-5: The Delivery Partner mobile application shall present pickup and drop-off details in a single-screen, large-touch-target layout suitable for one-handed use.

3.2 Hardware Interfaces

- HI-1: The Delivery Partner mobile application shall interface with the host device's GPS hardware to report real-time location for delivery tracking.
- HI-2: The Store Owner portal shall optionally interface with a connected barcode scanner (via USB or Bluetooth HID) to speed up product entry, where available.
- HI-3: No other dedicated hardware interfaces have been identified for Release 1.0.

3.3 Software Interfaces

Interface	Purpose	Protocol
Payment Gateway API	Process card, UPI, and wallet payments; issue refunds.	REST/HTTPS, PCI-DSS compliant
SMS Gateway	Deliver SMS notifications for order and OTP events.	REST/HTTPS
E-mail Service (e.g., SMTP/SendGrid)	Deliver transactional e-mails: verification, receipts, alerts.	SMTP / REST API
Push Notification Service (FCM / APNs)	Deliver mobile push notifications.	REST/HTTPS
Maps & Geolocation API	Distance-based search, delivery routing, address validation.	REST/HTTPS
Relational Database (PostgreSQL)	Persist users, stores, products, orders, and price history.	SQL over TLS

3.4 Communications Interfaces

- CI-1: All client-server communication shall occur over HTTPS using TLS 1.2 or higher.
- CI-2: The system shall use a WebSocket or equivalent push channel to deliver real-time price updates to an open product listing page without requiring a manual refresh.
- CI-3: The system shall send transactional e-mail messages to confirm order placement, order status changes, and refund completion.
- CI-4: The system shall send SMS messages for time-critical alerts, including one-time passwords and delivery-arrival notices.

4. System Features

This section itemizes the functional requirements of the system, organized by feature. Each feature includes a description and priority, representative stimulus/response sequences, and a numbered list of detailed functional requirements written in "the system shall" form. Requirement identifiers are prefixed with the feature's abbreviation (e.g., FR-URL for User Registration & Login) so that each requirement can be traced individually in a requirements-traceability matrix.

4.1 User Registration & Login

Description and Priority

This feature allows Customers, Grocery Store Owners, and Delivery Partners to create an account, verify their identity, and securely log in to access role-specific functionality of the marketplace.

Priority: High

Stimulus/Response Sequences

Stimulus: *A new user requests to register an account.*

Response: *The system collects the user's details, validates them, and creates an inactive account pending e-mail verification.*

Stimulus: *A registered user requests to log in with valid credentials.*

Response: *The system authenticates the user and grants access to the role-appropriate dashboard.*

Stimulus: *A user requests to reset a forgotten password.*

Response: *The system sends a time-limited password-reset link to the registered e-mail address.*

Functional Requirements

FR-URL-1: The system shall allow a new user to register with name, e-mail, phone number, password, and a selected role (Customer, Store Owner, Delivery Partner), validating e-mail/phone format and requiring e-mail verification before account activation.

FR-URL-2: The system shall enforce a minimum password strength (8+ characters with uppercase, numeral, and special character) and lock an account for 15 minutes after five consecutive failed login attempts.

FR-URL-3: The system shall authenticate a registered user via e-mail/phone and password, or via Google OAuth 2.0, and shall allow logout from any page.

FR-URL-4: The system shall provide a "Forgot Password" function that sends a time-limited, single-use password-reset link to the user's registered e-mail address.

4.2 Store Registration

Description and Priority

This feature allows a Grocery Store Owner to apply to list products on the marketplace and requires Administrator verification before the store may begin selling, per BR-1.

Priority: High

Stimulus/Response Sequences

Stimulus: *A Store Owner submits a store registration request.*

Response: *The system records the request with status "Pending Verification" and notifies the Administrator.*

Stimulus: *The Administrator approves or rejects a pending store registration.*

Response: *The system updates the store status accordingly and notifies the Store Owner.*

Functional Requirements

FR-SR-1: The system shall allow a Store Owner to submit a store registration request (store name, address, GST/tax ID, contact details, and business-license upload) and shall set its initial status to "Pending Verification."

FR-SR-2: The system shall notify the Administrator of new registration requests and allow the Administrator to approve or reject each request, per BR-1, notifying the Store Owner of the outcome.

FR-SR-3: The system shall allow an approved Store Owner to update the store profile (hours, pickup address, delivery radius) and shall allow the Administrator to suspend or reinstate a store account, per BR-8.

4.3 Product Listing

Description and Priority

This feature allows a verified Store Owner to publish surplus or near-expiry products for sale, capturing the information the pricing and expiry-tracking engines depend on.

Priority: High

Stimulus/Response Sequences

Stimulus: *A Store Owner requests to create a new product listing.*

Response: *The system prompts for product details, including expiry date and time, and publishes the listing once validated.*

Stimulus: *A Store Owner requests to edit or deactivate an existing listing.*

Response: *The system applies the change, subject to whether the item has already been sold.*

Functional Requirements

FR-PL-1: The system shall allow a Store Owner to create a listing with name, category, description, up to five images, original price, quantity, and expiry date/time, rejecting any expiry date/time at or before the current time, per BR-4.

FR-PL-2: The system shall allow the Store Owner to edit a listing's description, images, or quantity before its first sale, and to manually deactivate a listing at any time.

FR-PL-3: The system shall display, for every active listing, both the original price and the current dynamically calculated price, together with the hours remaining until expiry.

4.4 Inventory Management

Description and Priority

This feature maintains an accurate, real-time count of available product units for each store so that customers cannot purchase more than is physically in stock.

Priority: High

Stimulus/Response Sequences

Stimulus: *A Customer's order is confirmed with payment.*

Response: *The system decrements the available quantity of each ordered product.*

Stimulus: *A Store Owner uploads a bulk inventory file.*

Response: *The system validates and imports each row, reporting any errors.*

Functional Requirements

FR-IM-1: The system shall maintain a real-time inventory count per product, automatically decrementing it on confirmed order and incrementing it on cancellation or refund.

FR-IM-2: The system shall alert the Store Owner when a product's quantity reaches zero and shall exclude zero-quantity products from active search results.

FR-IM-3: The system shall allow the Store Owner to bulk-upload inventory via CSV (name, quantity, price, category, expiry date/time), validating each row and reporting errors by row number.

4.5 Expiry Tracking

Description and Priority

This feature continuously monitors the remaining shelf life of every listed product, classifies it into an expiry-proximity band, and removes it from sale once expired.

Priority: High

Stimulus/Response Sequences

Stimulus: *The scheduled expiry-tracking job runs (hourly).*

Response: *The system recalculates the expiry-proximity band of every active listing and flags critical items.*

Stimulus: *A product's expiry date and time is reached.*

Response: *The system automatically removes the listing from the marketplace.*

Functional Requirements

FR-ET-1: The system shall recalculate the expiry-proximity band (per BR-5) of every active listing at least once every hour and shall flag a listing as "Critical" when fewer than 6 hours remain.

FR-ET-2: The system shall notify the Store Owner at 24, 6, and 1 hour before an unsold listing's expiry.

FR-ET-3: The system shall automatically and permanently remove a listing from the marketplace at its recorded expiry date/time, per BR-4, and shall log the removal with a timestamp.

4.6 Dynamic Hourly Pricing Engine

Description and Priority

This is the core feature of the marketplace: an automated engine that recalculates each product's selling price every hour as it approaches expiry, applying a progressively steeper discount, per BR-2 and BR-3.

Priority: High

Stimulus/Response Sequences

Stimulus: *The scheduled hourly pricing job runs.*

Response: *The system recalculates and publishes an updated price for every active product listing.*

Stimulus: *A Store Owner manually edits a product's base price or discount configuration.*

Response: *The system immediately recalculates that product's current price.*

Functional Requirements

FR-DPE-1: The system shall recalculate the price of every active listing once every hour from its base price, the configured decay rate, and hours remaining to expiry, per BR-2.

FR-DPE-2: The system shall not discount a listing beyond 80 percent of its original price nor below its configured minimum price floor, per BR-3.

FR-DPE-3: The system shall allow the Store Owner to configure, per category, the decay curve (linear or stepped) and the hour offset at which decay pricing begins.

FR-DPE-4: The system shall record a timestamped price-history entry for every hourly change and shall reflect updated prices on the storefront within 5 minutes.

FR-DPE-5: The system shall immediately recalculate a listing's price whenever its base price or discount configuration is manually edited, without waiting for the next hourly cycle.

4.7 Product Search & Filters

Description and Priority

This feature helps Customers quickly locate relevant discounted, near-expiry products from among the stores available to them.

Priority: Medium

Stimulus/Response Sequences

Stimulus: *A Customer enters a search keyword or applies filters.*

Response: *The system returns matching active, in-stock, non-expired listings ranked per the selected sort order.*

Functional Requirements

FR-PSF-1: The system shall allow a Customer to search by keyword, category, or store name, with autocomplete suggestions, returning only active, in-stock, non-expired listings.

FR-PSF-2: The system shall allow filtering by discount percentage, price range, expiry-proximity band, and distance, and sorting by highest discount, lowest price, or nearest expiry.

4.8 Shopping Cart

Description and Priority

This feature lets a Customer collect items from one or more stores before checkout, while protecting against the price changing mid-session due to the hourly pricing engine.

Priority: High

Stimulus/Response Sequences

Stimulus: *A Customer requests to add a product to the cart.*

Response: *The system adds the item, subject to available inventory, and locks its price for the holding period.*

Stimulus: *The 15-minute cart price-holding period elapses before checkout.*

Response: *The system refreshes the item's price and notifies the Customer.*

Functional Requirements

FR-SC-1: The system shall allow a Customer to add, adjust the quantity of, or remove a cart item (subject to available inventory) and shall lock each item's price for 15 minutes from addition, per BR-7.

FR-SC-2: The system shall refresh and notify the Customer of any cart item whose price hold has expired, or that has expired/been removed from the marketplace, before permitting checkout.

FR-SC-3: The system shall display the subtotal, delivery fee, and total for the cart, and shall support items from multiple stores by grouping them into separate orders per store at checkout.

4.9 Secure Payment

Description and Priority

This feature processes Customer payments through a PCI-DSS compliant third-party payment gateway and manages refunds for cancelled or disputed orders.

Priority: High

Stimulus/Response Sequences

Stimulus: *A Customer confirms checkout and selects a payment method.*

Response: *The system routes the transaction to the payment gateway and displays the confirmation result.*

Stimulus: *A refund is approved for a cancelled or disputed order.*

Response: *The system initiates a refund to the Customer's original payment method.*

Functional Requirements

FR-SP-1: The system shall process payments (credit/debit card, UPI, digital wallets) through a PCI-DSS compliant gateway, encrypting all payment data in transit with TLS 1.2+, and shall not store raw card numbers.

FR-SP-2: The system shall display a confirmation and send an e-mail receipt with a unique transaction ID on successful payment, and shall offer retry or an alternate method on failure.

FR-SP-3: The system shall process approved refunds to the Customer's original payment method within 5-7 business days, per BR-6.

4.10 Order Management

Description and Priority

This feature governs the lifecycle of an order from placement through delivery, coordinating the Customer, Store Owner, and Delivery Partner.

Priority: High

Stimulus/Response Sequences

Stimulus: *A Customer completes checkout with successful payment.*

Response: *The system creates one order per store, notifies the relevant Store Owner, and sets order status to "Placed."*

Stimulus: *A Store Owner does not respond to a new order within 15 minutes.*

Response: *The system automatically cancels the order and refunds the Customer, per BR-6.*

Functional Requirements

FR-OM-1: The system shall allow a Customer to place an order for the items in the shopping cart following successful payment.

FR-OM-2: The system shall assign a unique, sequential order number to every accepted order.

FR-OM-3: The system shall set the initial status of a new order to "Placed" and shall progress it through "Confirmed," "Preparing," "Out for Delivery," "Delivered," or "Cancelled," per the state-transition model in Appendix A.

FR-OM-4: The system shall notify the relevant Store Owner immediately upon creation of a new order.

FR-OM-5: The system shall allow the Store Owner to accept or reject an order within 15 minutes of receipt.

FR-OM-6: The system shall automatically cancel an order and initiate a refund if the Store Owner does not respond within 15 minutes, per BR-6.

FR-OM-7: The system shall allow a Customer to cancel an order only while its status is "Placed" or "Confirmed."

FR-OM-8: The system shall allow a Delivery Partner to accept an assigned delivery request and update its delivery status.

FR-OM-9: The system shall allow a Customer to view the order history and current status of all past and active orders.

FR-OM-10: The system shall allow a Customer to rate the order and the Delivery Partner after delivery is marked complete.

4.11 Notifications

Description and Priority

This feature keeps Customers and Store Owners informed of time-sensitive events such as order updates, price drops, and impending expiry.

Priority: Medium

Stimulus/Response Sequences

Stimulus: *An order status changes, or a wishlisted item's price drops.*

Response: *The system sends a push notification and/or e-mail to the affected Customer per their notification preferences.*

Functional Requirements

FR-NOT-1: The system shall send push notifications and e-mail alerts to Customers for order confirmation, order status changes, and delivery updates.

FR-NOT-2: The system shall notify Customers of newly listed near-expiry products that match their saved preferences or wishlist.

FR-NOT-3: The system shall notify Store Owners of low-inventory conditions and impending product expiry.

FR-NOT-4: The system shall allow a user to configure notification channels (push, e-mail, SMS) from the account settings page.

FR-NOT-5: The system shall notify a Customer when the price of a wishlisted item drops as a result of the hourly decay-pricing recalculation.

FR-NOT-6: The system shall send an automated notification to the Administrator when a store's order-rejection rate exceeds the threshold defined in BR-8.

4.12 Food Waste Analytics Dashboard

Description and Priority

This feature gives Store Owners and the Administrator visibility into how much food has been sold versus wasted, and the estimated impact of the decay-pricing model.

Priority: Medium

Stimulus/Response Sequences

Stimulus: *A Store Owner opens the analytics dashboard.*

Response: *The system displays sales-versus-waste metrics for the selected time period.*

Functional Requirements

FR-FWA-1: The system shall display, per selectable time period and category, the quantity of food sold versus wasted, the average discount at which items sold, and the estimated monetary value of waste avoided.

FR-FWA-2: The system shall allow the Store Owner to export analytics data (CSV/PDF) and shall provide the Administrator an aggregated, anonymized platform-wide waste-reduction metric, refreshed at least every 24 hours.

4.13 Admin Panel

Gives the Administrator the tools to govern the marketplace: approving stores, moderating listings, configuring platform parameters, and monitoring health.

Stimulus: Administrator reviews a pending approval, dispute, or configuration change. → Response: System applies the action and records it in the audit log.

FR-AP-1: The system shall allow the Administrator to approve, reject, suspend, and reinstate store, Customer, and Delivery Partner accounts.

FR-AP-2: The system shall allow the Administrator to configure platform-wide parameters (e.g., discount cap, cart holding period, category list) per BR-3 and BR-7, and to resolve disputes and view platform reports.

FR-AP-3: The system shall allow the Administrator to manually remove a policy-violating listing and shall maintain an audit log of all administrative actions with timestamps.

5. Other Non-Functional Requirements

This section states the quality attributes the system must exhibit, independent of any single feature. Rationale is provided where it helps clarify design intent.

5.1 Performance Requirements

- PE-1: The system shall support at least 5,000 concurrent active users during peak hours (7:00 PM to 10:00 PM local time) without degradation in response time.
- PE-2: 95 percent of product search queries shall return results within 2 seconds under normal load.
- PE-3: The hourly pricing recalculation batch job shall complete for all active listings within 10 minutes of being triggered.
- PE-4: Checkout and payment confirmation shall complete within 7 seconds of the Customer submitting payment, exclusive of pyment-gateway processing time.
- PE-5: All customer-facing web pages shall be fully rendered within 3 seconds on a 4G mobile connection.

5.2 Reliability Requirements

- RE-1: The system shall have a Mean Time Between Failures (MTBF) of at least 720 hours for core transactional services (order placement and payment).
- RE-2: The pricing engine shall log and alert on any hourly recalculation cycle that fails to complete, and shall automatically retry the failed cycle within 5 minutes.
- RE-3: The system shall preserve cart contents and in-progress orders through a transient service failure, and shall allow the user to resume without data loss, consistent with the Robustness principle applied in the Cafeteria Ordering System precedent.

5.3 Availability Requirements

- AV-1: The system shall be available 99.5 percent of the time, measured monthly, excluding scheduled maintenance windows.
- AV-2: Scheduled maintenance shall be performed during off-peak hours (2:00 AM to 4:00 AM local time) and announced to users at least 24 hours in advance.
- AV-3: The pricing engine and inventory service shall be deployed with automatic failover such that a single node failure does not interrupt hourly price recalculation.

5.4 Scalability Requirements

- SC-NFR-1: The system architecture shall support horizontal scaling of the pricing engine and search service independently of other services to accommodate growth in the number of listed products.
- SC-NFR-2: The system shall support onboarding at least 2,000 partner stores and 500,000 registered Customers without requiring architectural redesign.
- SC-NFR-3: The database schema and indexing strategy shall support at least 1 million active product listings with search response times remaining within the limits specified in PE-2.

5.5 Maintainability Requirements

- MA-1: The system shall be composed of loosely coupled microservices such that the pricing algorithm can be modified or replaced without requiring changes to the order or payment services.
- MA-2: All business-rule constants (e.g., maximum discount cap, cart holding period, order-rejection timeout) shall be externally configurable by the Administrator without requiring a code deployment.
- MA-3: The codebase shall maintain a minimum of 80 percent automated test coverage for the pricing engine and payment modules.

5.6 Security Requirements

- SE-1: All network transactions involving financial or personally identifiable information shall be encrypted in transit using TLS 1.2 or higher, per BR and Section 3.4.
- SE-2: Users shall be required to log in to the system for all operations except browsing publicly listed products.
- SE-3: Passwords shall be stored using a salted, industry-standard one-way hashing algorithm (e.g., bcrypt or Argon2); plaintext passwords shall never be stored or logged.
- SE-4: The system shall permit only Administrators to approve, reject, suspend, or reinstate store accounts.
- SE-5: The system shall permit a Customer to view only their own order history, not orders placed by other Customers.
- SE-6: The system shall implement role-based access control such that a Store Owner can view and manage only their own store's listings, inventory, and orders.

5.7 Privacy Requirements

- PR-1: The system shall collect only the personal data necessary to fulfill registration, ordering, and delivery, consistent with the Digital Personal Data Protection Act, 2023 [4].
- PR-2: The system shall allow a Customer to request deletion of their personal account data, subject to statutory record-retention requirements described in Section 7.
- PR-3: The system shall not share a Customer's personal contact information with a Store Owner or Delivery Partner beyond what is strictly necessary to fulfill a specific order.
- PR-4: The system shall obtain explicit user consent before sending marketing (non-transactional) notifications.

5.8 Usability Requirements

- US-1: A first-time Customer shall be able to complete registration and place a first order within 5 minutes without external assistance.
- US-2: The system shall conform to WCAG 2.1 Level AA accessibility guidelines for color contrast, text scaling, and screen-reader compatibility.
- US-3: The system shall support English and Hindi language interfaces at Release 1.0, with the language selectable from account settings.

5.9 Portability Requirements

- PO-1: The customer-facing and store-owner web applications shall render correctly across the browsers listed in Section 2.4 (OE-1) without requiring browser-specific code paths.
- PO-2: The mobile applications shall share a common backend API layer such that no platform-specific business logic resides in the Android or iOS client.
- PO-3: The system's backend services shall be containerized to allow deployment on any Kubernetes-compatible cloud infrastructure.

6. Business Rules

Business rules state the platform's operating principles; the functional requirements above implement and enforce them.

ID	Rule	Statement
BR-1	Store Verification	A store shall not list products until the Administrator has verified its business license/GST registration and approved its registration.
BR-2	Hourly Discount Calculation	Each listing's discount is recalculated hourly from its base price, store-configured decay rate, and hours remaining to expiry; perishable categories (e.g., Dairy, Bakery) default to a steeper decay curve than shelf-stable ones unless overridden.
BR-3	Maximum Discount & Price Floor	Cumulative discount shall never exceed 80% of the original price, and shall never go below the Store Owner's configured minimum price floor for that category.
BR-4	Product Expiry & Auto-Removal	A listing with an expiry date/time at or before the current time shall be rejected at creation; an active listing shall be automatically and permanently removed the moment its expiry is reached.
BR-5	Expiry-Proximity Bands	Listings are classified as Critical (0-6h), Urgent (6-24h), Soon (1-3 days), or Standard (3+ days), driving both pricing decay and customer-facing filters.
BR-6	Order Timeout & Refund Policy	A Store Owner who does not act on an order within 15 minutes has it auto-cancelled and refunded; refunds (for cancellation, spoiled/late delivery, or dispute) are processed within 5-7 business days.
BR-7	Cart Price Holding Period	An item's price is locked for 15 minutes from being added to cart, after which it refreshes to the current computed price.
BR-8	Store Suspension Threshold	The Administrator is auto-notified when a store's 30-day rolling order-rejection rate exceeds 20%, and may suspend its listing privileges pending review.

7. Other Requirements

7.1 Legal and Regulatory Requirements

- OR-1: The system shall retain order and transaction records for a minimum of 3 years to satisfy applicable Indian tax and audit requirements.
- OR-2: The system shall display FSSAI-compliant food safety and labelling information on the product detail page for applicable packaged food categories.
- OR-3: The system shall provide a mechanism for a Customer to exercise data-access and data-deletion rights under the Digital Personal Data Protection Act, 2023 [4].

7.2 Internationalization Requirements

- OR-4: The system shall display currency values in Indian Rupees (INR) with the ₹ symbol at Release 1.0.

- OR-5: The system's date and time displays shall use the Indian Standard Time (IST) zone unless a future release introduces multi-region operation.

7.3 Reporting Requirements

- OR-6: The system shall generate a monthly platform-wide summary report of total orders, total food-waste value avoided, and active store count for internal business review.

8. Glossary

Term	Definition
Decay Pricing	Automated, progressive price reduction as a product's expiry approaches.
Expiry-Proximity Band	Classification (Critical/Urgent/Soon/Standard) based on time remaining to expiry, per BR-5.
Price Floor	The minimum price below which the pricing engine will not discount an item, per BR-3.
Surplus Inventory	Store stock in excess of near-term demand, listed on the marketplace at a discount.
Cart Price Holding Period	The 15-minute window during which an added item's price is locked, per BR-7.
Order Rejection Timeout	The 15-minute window for a Store Owner to act on a new order before auto-cancellation, per BR-6.
Waste Avoided	Inventory sold at any discount before expiry, rather than expiring unsold.
PCI-DSS	Payment Card Industry Data Security Standard for organizations handling card data.
UPI	Unified Payments Interface; a real-time India-based interbank payment system.
DPDP Act	India's Digital Personal Data Protection Act, 2023, governing personal-data processing.

Appendix A: Analysis Models

This appendix describes the analysis models used to design the system: the use case diagram, context diagram, Level-0 data flow diagram, entity-relationship model, class diagram, activity flow, and product-lifecycle state diagram. These descriptions are provided in structured textual form so that they can be directly translated into formal diagrams (e.g., in UML or DFD notation) during design.

A.1 Use Case Diagram Description

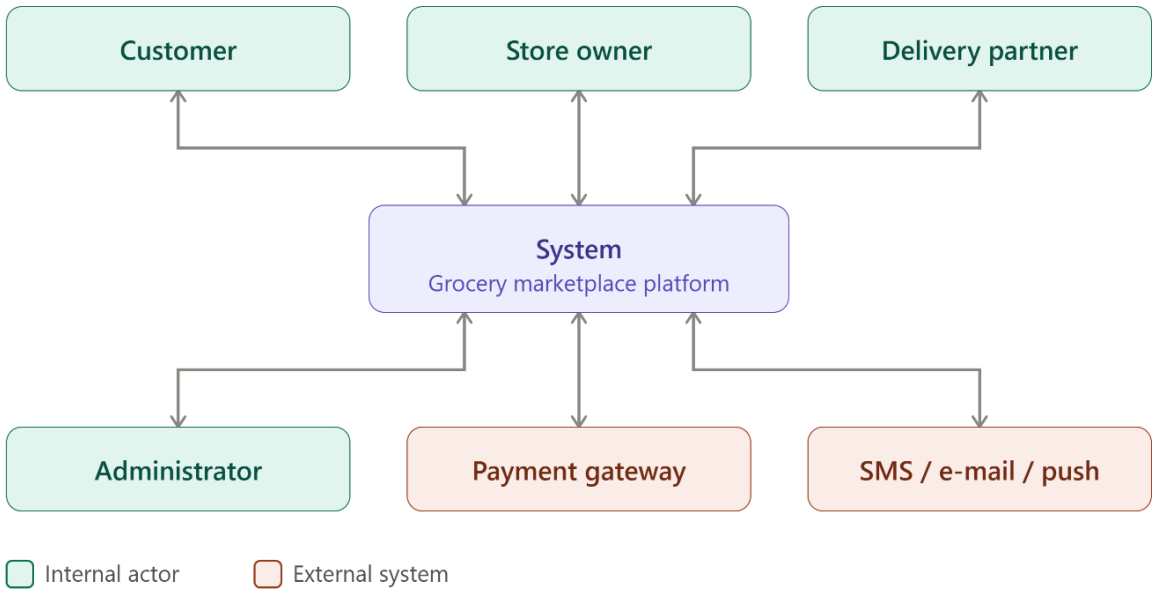
The system has five actors: Customer, Grocery Store Owner, Delivery Partner, Administrator, and the external Payment Gateway. Principal use cases per actor are summarized below.

Actor	Principal Use Cases
Customer	Register/Login, Search Products, Add to Cart, Checkout, Make Payment, Track Order, Rate Order, Manage Wishlist, View Notifications
Grocery Store Owner	Register Store, Create Product Listing, Manage Inventory, Configure Pricing Rules, Accept/Reject Order, View Analytics Dashboard
Delivery Partner	Accept Delivery Request, View Pickup/Drop-off Details, Update Delivery Status, View Earnings
Administrator	Approve/Reject Store Registration, Suspend Account, Configure Business Rules, Resolve Disputes, View Platform Reports
Payment Gateway (external)	Authorize Payment, Process Refund, Send Transaction Status

A.2 Context Diagram Description

The Dynamic Expiry-Based Grocery & Zero-Waste Marketplace forms the center of the context diagram, exchanging the following data flows with six external entities:

- Customer ⇄ System: search queries, cart/order/payment data (outbound); product listings, order status, notifications (inbound).
- Grocery Store Owner ⇄ System: store registration, product listings, pricing configuration (outbound); order notifications, analytics data (inbound).
- Delivery Partner ⇄ System: delivery status updates, location data (outbound); delivery assignment details (inbound).
- Administrator ⇄ System: verification decisions, business-rule configuration (outbound); pending approvals, dispute tickets, platform reports (inbound).
- Payment Gateway ⇄ System: payment authorization requests, refund requests (outbound); transaction status responses (inbound).
- SMS/E-mail/Push Notification Services ⇄ System: notification dispatch requests (outbound); delivery receipts (inbound).



A.3 DFD Level 0 Description

The Level-0 data flow diagram decomposes the system into five major processes and four persistent data stores:

Process	Description
1.0 Manage Users & Stores	Handles registration, authentication, and store verification.
2.0 Manage Listings & Pricing	Handles product listing, inventory updates, expiry tracking, and the hourly pricing engine.
3.0 Process Orders & Payments	Handles cart, checkout, payment authorization, and refunds.
4.0 Manage Delivery	Handles delivery assignment and status tracking.

Process	Description
5.0 Generate Analytics	Aggregates sales and waste data into the food-waste analytics dashboard.

Data stores: D1 User/Store Store, D2 Product/Inventory Store, D3 Order/Payment Store, D4 Price-History Store. Process 2.0 reads from and writes to D2 and D4 every hour; Process 3.0 reads from D2 to validate inventory and writes to D3 on order placement; Process 5.0 reads from D2, D3, and D4 to compute waste-avoidance metrics.

A.4 ER Diagram Entities

Entity	Key Attributes	Relationships
User	user_id, name, email, phone, role, password_hash	1 User places many Orders; 1 User owns 0..1 Store
Store	store_id, owner_id, name, address, gst_number, status	1 Store lists many Products; verified by Administrator
Product	product_id, store_id, name, category, base_price, expiry_datetime, quantity	1 Product has many PriceHistory entries; belongs to 1 Store
PriceHistory	history_id, product_id, price, calculated_at	Many PriceHistory rows belong to 1 Product
Order	order_id, customer_id, store_id, status, total_amount, placed_at	1 Order has many OrderItems; placed by 1 Customer
OrderItem	item_id, order_id, product_id, quantity, price_at_purchase	Many OrderItems belong to 1 Order and reference 1 Product
Payment	payment_id, order_id, amount, method, status, transaction_id	1 Payment belongs to 1 Order
DeliveryAssignment	assignment_id, order_id, delivery_partner_id, status	1 DeliveryAssignment belongs to 1 Order
Notification	notification_id, user_id, type, message, sent_at	Many Notifications belong to 1 User
Review	review_id, order_id, customer_id, rating, comment	1 Review belongs to 1 Order

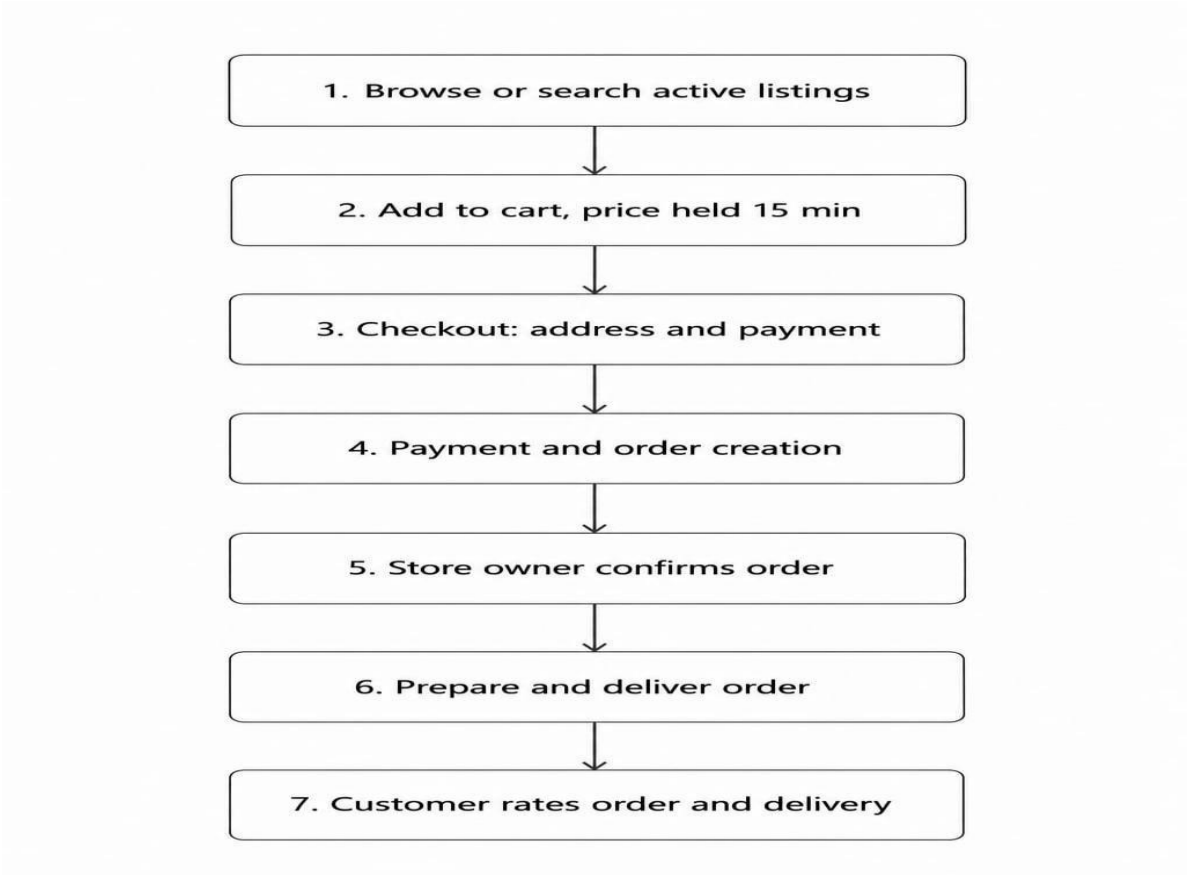
A.5 Class Diagram Classes

Class	Key Responsibilities
User (abstract)	Common authentication attributes; subclassed by Customer, StoreOwner, DeliveryPartner, Administrator.
Store	Holds store profile and verification status; aggregates Product listings.
Product	Holds listing details and current price; exposes calculateCurrentPrice().
PricingEngine	Executes the hourly decay-pricing calculation across all active Products.
InventoryManager	Tracks and updates available quantity per Product.
Cart / CartItem	Holds a Customer's in-progress selections with price-hold logic.
Order / OrderItem	Represents a placed order and its line items; manages status transitions.
Payment	Encapsulates payment method, status, and gateway transaction reference.

Class	Key Responsibilities
NotificationService	Dispatches push/e-mail/SMS notifications based on user preferences.
AnalyticsReport	Aggregates sales-versus-waste data for the dashboard.

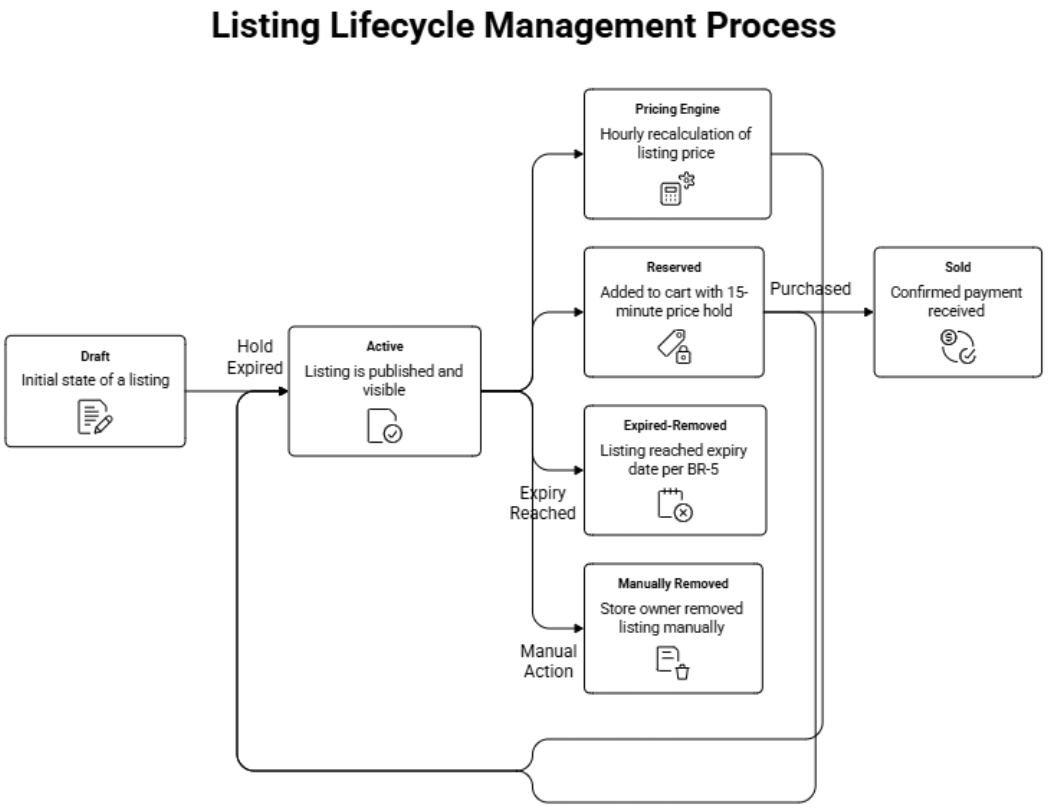
A.6 Activity Diagram Flow — "Place Order"

The principal customer-facing activity flow proceeds as follows:



A.7 State Diagram for Product Lifecycle

Every product listing progresses through a well-defined set of states from creation to sale, expiry, or removal, as shown in Figure A-1.



As shown above: a listing begins as Draft, moves to Active upon publication, and is recalculated hourly by the pricing engine while Active. When a Customer adds it to the cart it becomes Reserved (with the price held for 15 minutes); it returns to Active if the hold expires unpurchased, or becomes Sold upon confirmed payment. An Active listing that reaches its expiry date and time without being sold automatically becomes Expired-Removed, per BR-5; a Store Owner may also manually transition an Active listing to Manually Removed at any time.